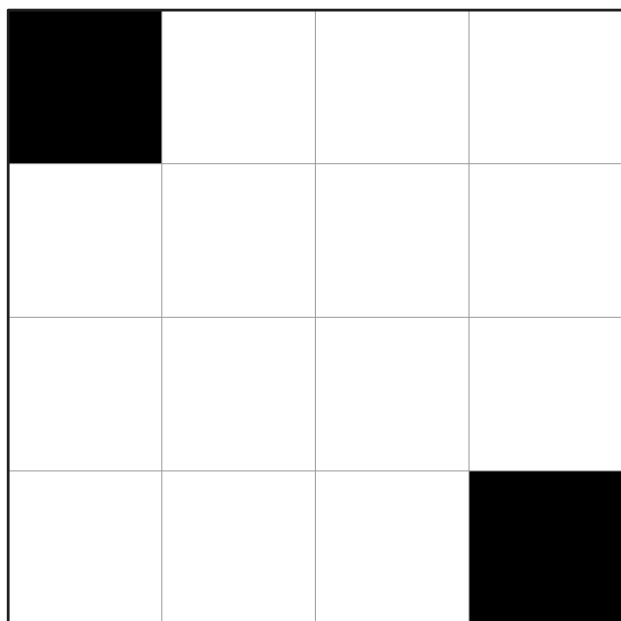
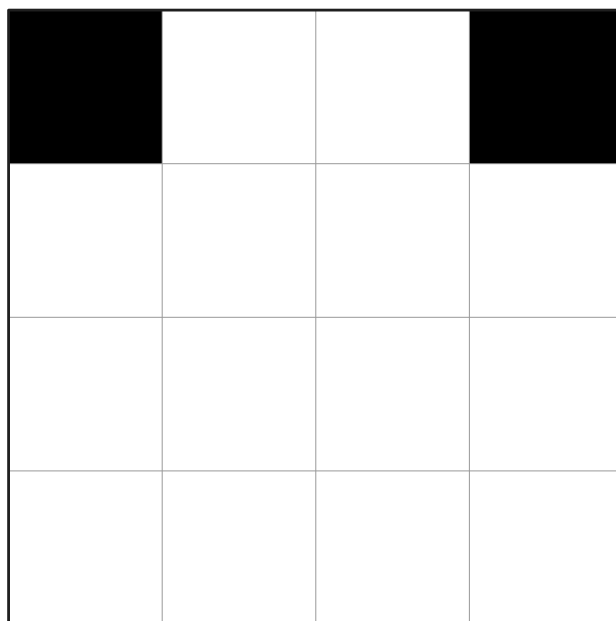


Problem 1: A domino covers two squares that share a side. On each of the boards A, B, C and D the two black squares may not be covered. Cover the 14 white squares with 7 dominoes: no gaps, no overlaps, nothing sticking out. When you succeed, draw the dominoes on the board (one thick line through the two squares of each domino) and write Y under it. When you have failed from at least three different starts, write N under it.

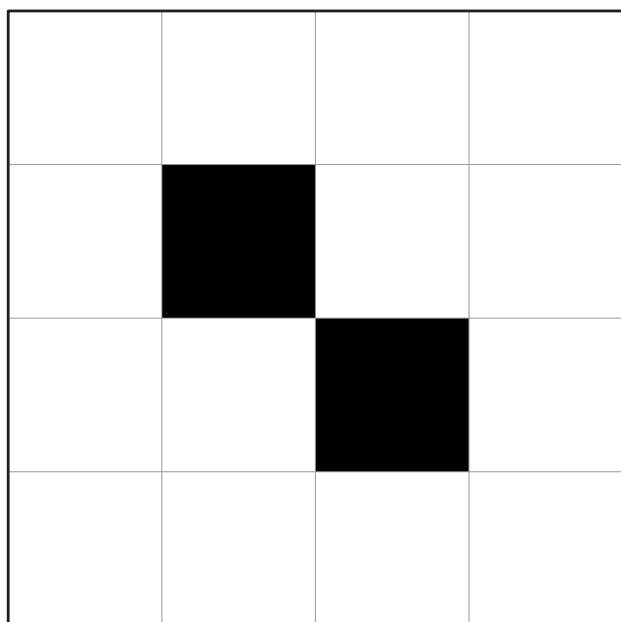
Board A



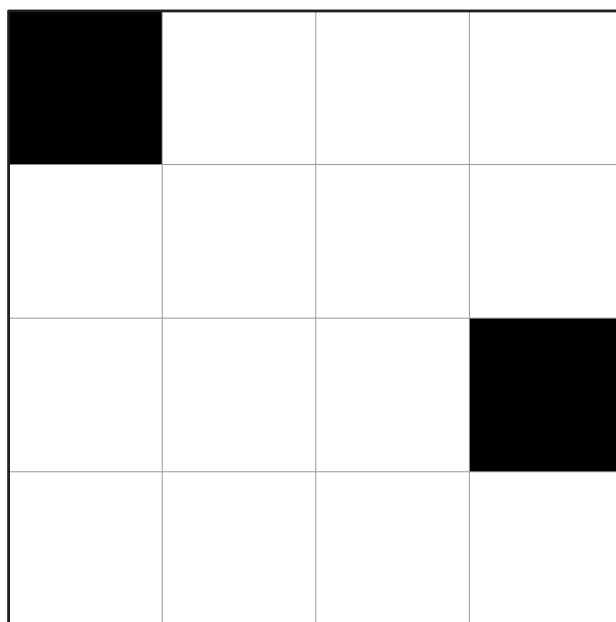
Board B



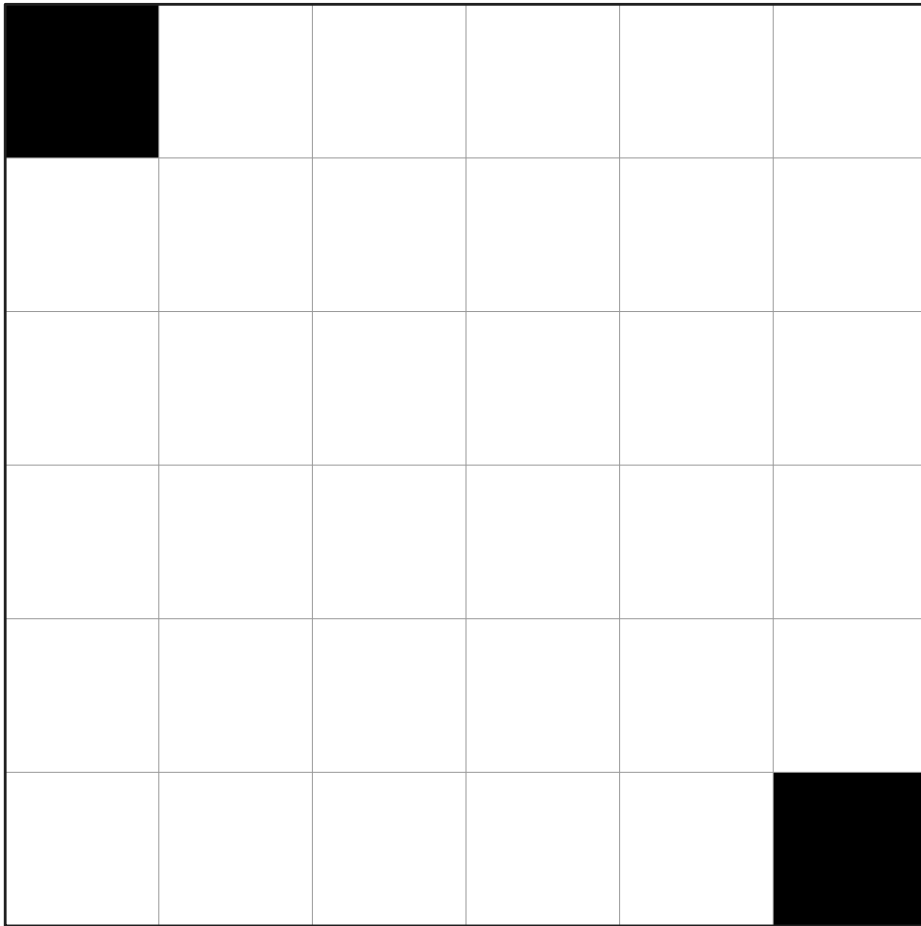
Board C



Board D



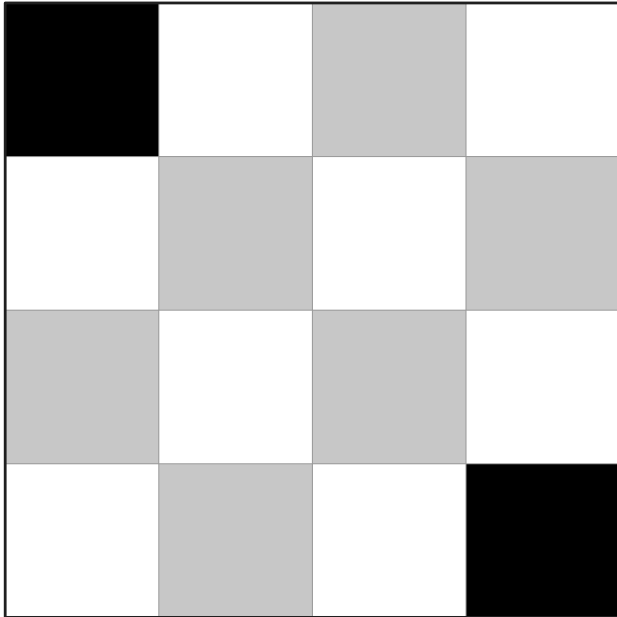
Problem 2: This 6 by 6 board has two opposite corners black. Cover the 34 white squares with 17 dominoes. Try for three minutes, changing your starting corner at least once. Write Y or N, and the largest number of dominoes you managed to place before getting stuck.



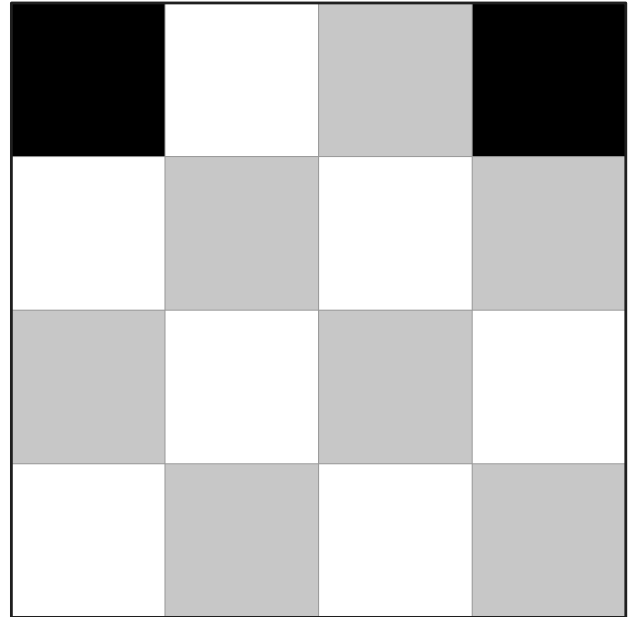
Y or N: _____

most dominoes
placed: _____

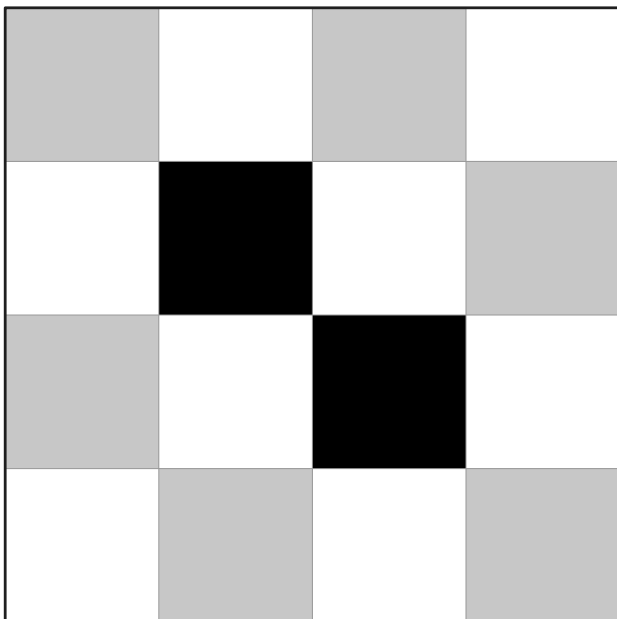
Problem 3: Here are boards A, B, C and D coloured like a checkerboard. Put a domino anywhere on one of them and look at the two squares under it: how many are dark and how many are light? Check five different places. Then, for each board, count the dark squares and the light squares that are left to cover, and write the two numbers under the board next to your Y or N from Problem 1. Write in one sentence what the N boards have in common.



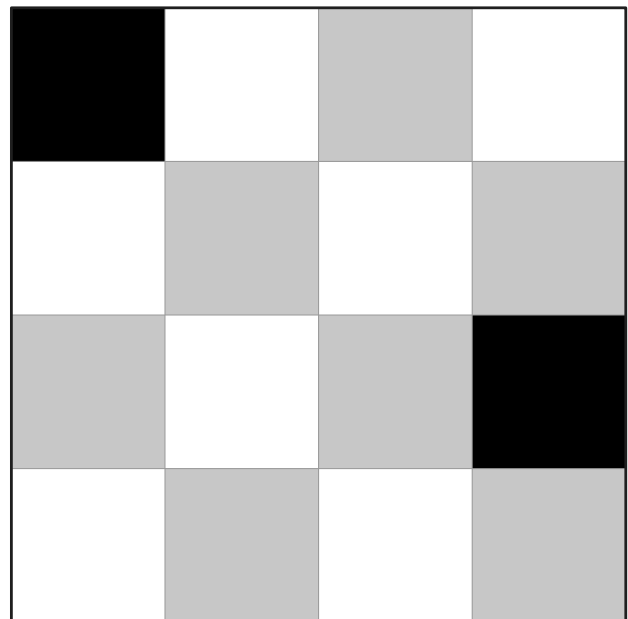
A: dark ____ light ____ Y or N ____



B: dark ____ light ____ Y or N ____



C: dark ____ light ____ Y or N ____

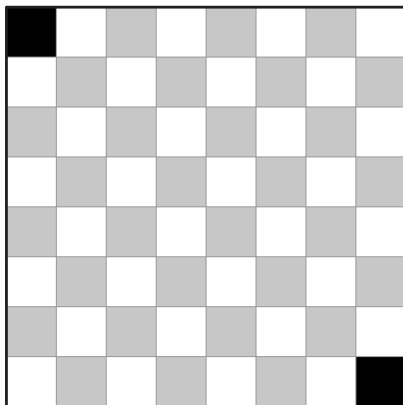
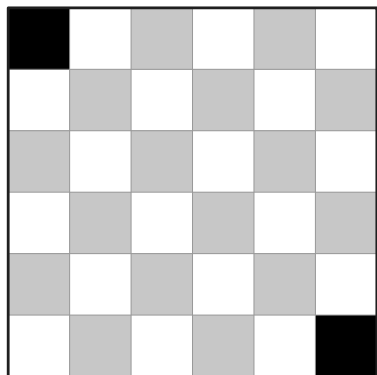


D: dark ____ light ____ Y or N ____

The N boards: _____

Problem 4: The 6 by 6 board from Problem 2, coloured the same way: count the dark squares and the light squares left to cover. Now the 8 by 8 board with two opposite corners removed: 62 squares, 31 dominoes. Decide Y or N without building, and write the reason in one or two sentences using the two counts.

6 by 6



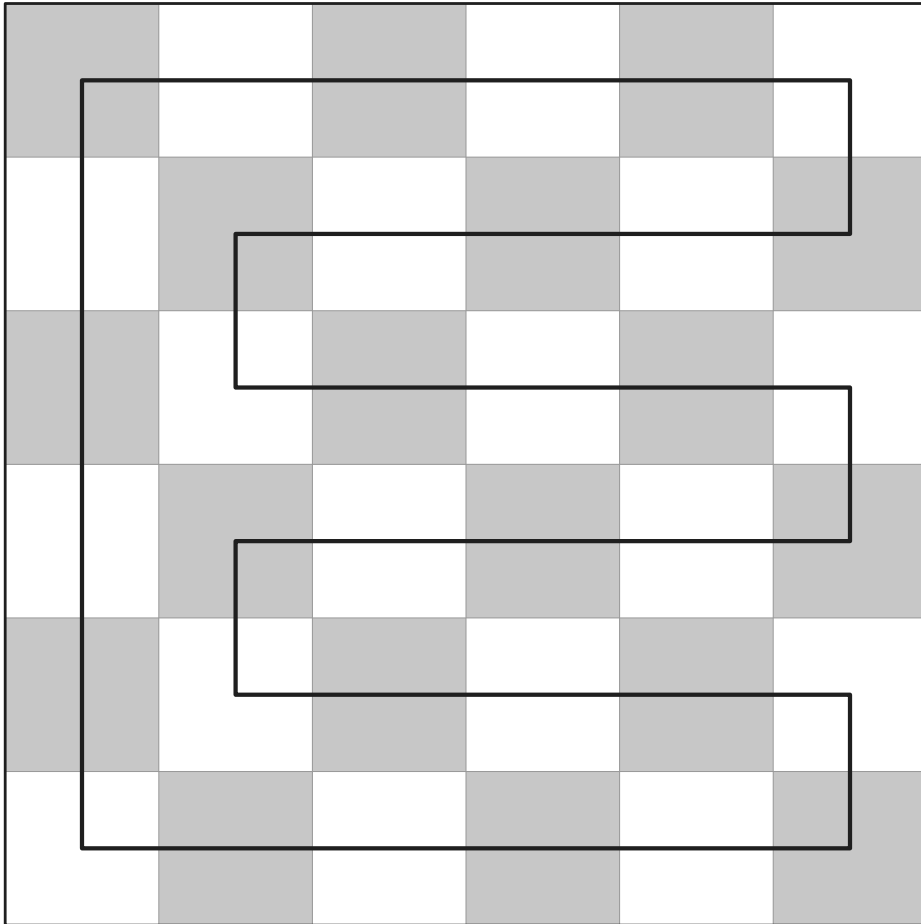
6 by 6: dark _____ light _____

8 by 8: dark _____ light _____

8 by 8, Y or N: _____

because _____

Problem 5: A path goes through all 36 squares of this board and returns to its start. Choose one dark square and one light square anywhere and put the two black paper squares on them. The path is cut into two roads (a road may have no squares at all). Count the squares on each road and write the two numbers. Then lay dominoes along each road, two squares at a time. Do this three times; the second time choose two squares far apart. Then write in one sentence why every pair of one dark and one light square can be done this way.



first pair:

roads of ____ and ____

second pair:

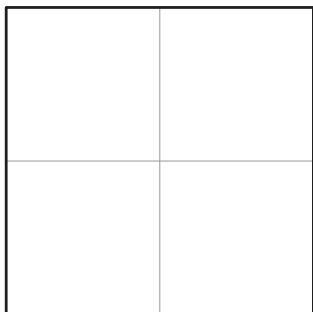
roads of ____ and ____

third pair:

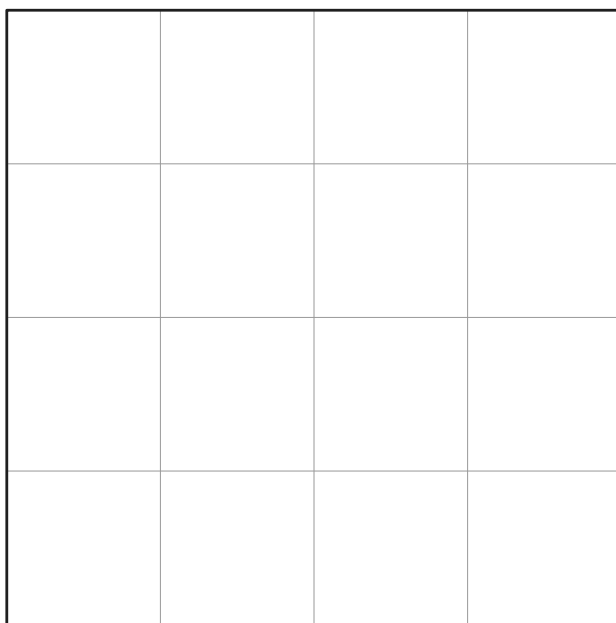
roads of ____ and ____

Every dark and light pair works because _____

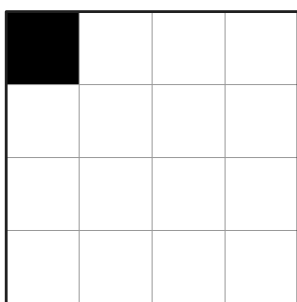
Problem 6: An L covers three squares in an L shape. It may be turned and flipped over. Put the black paper square on any square of the 2 by 2 board and cover the other three squares with one L (one piece counts as a covering). Move the black square to each of the other three places and cover again each time.



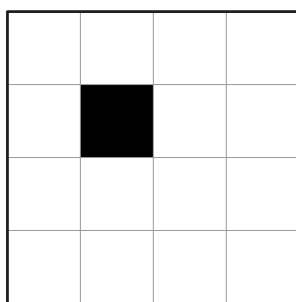
Problem 7: Put the black paper square on the big board at the square shown in picture P. Cover the other 15 squares with five L's, and colour the five L's on picture P. Do the same with the black square at the square shown in Q, then in R. Then choose any square you like, colour it black on S, and cover the rest.



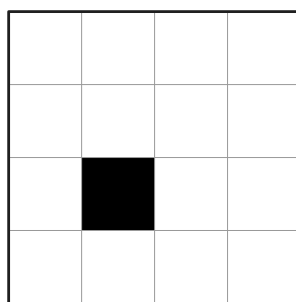
P



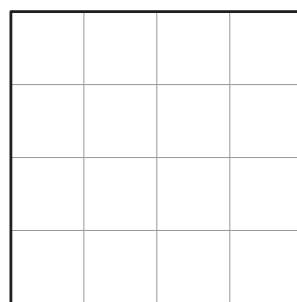
Q



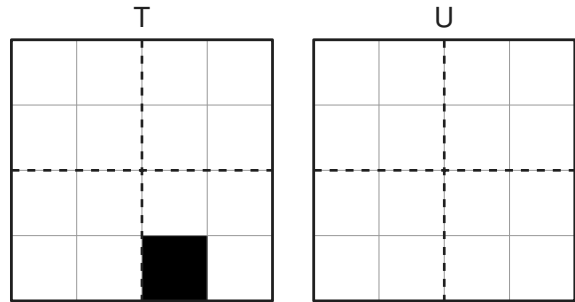
R



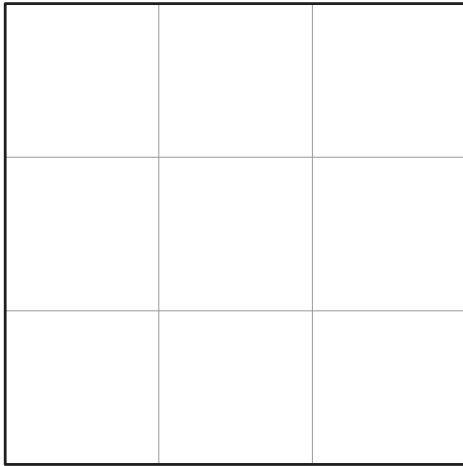
S



Problem 8: A recipe for the black square anywhere: draw the two middle lines of the board, cutting it into four 2 by 2 quarters. One quarter holds the black square. Put one L in the very middle of the board, covering one square of each of the other three quarters. Now every quarter has exactly one square that is black or covered; cover its other three squares with one L. Follow the recipe on picture T and colour the five L's. Then colour a black square of your own choice on U and follow the recipe again.

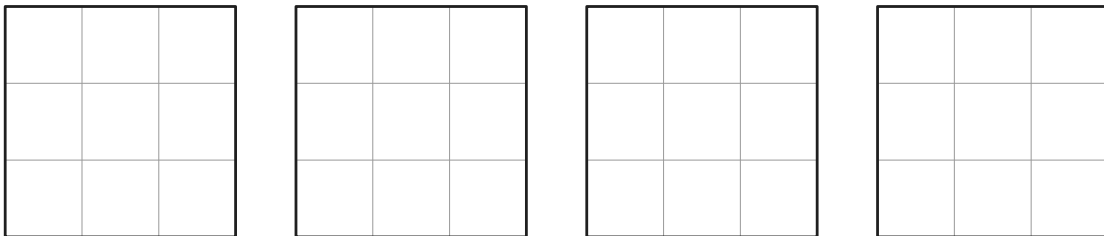


Problem 9: Try to cover the 3 by 3 board with three L's. Can you? Write yes or no.



yes or no: _____

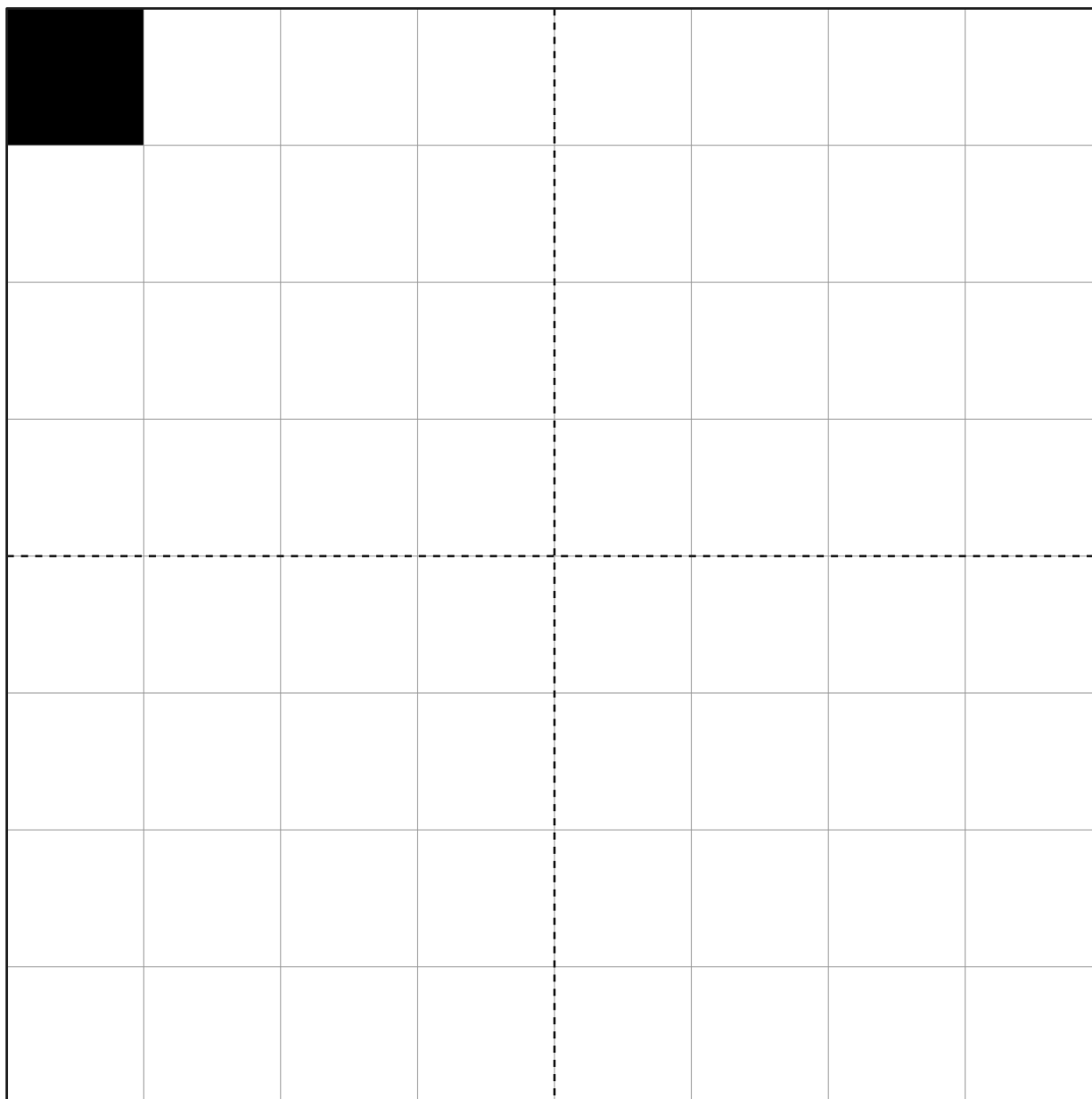
Problem 10: Put one L on the 3 by 3 board so that it covers the top-left corner square. Colour that position on a small board. Find every position that covers the top-left corner square and colour each one on its own small board. Does an L ever cover two corner squares at the same time? Count the corner squares and count the L's. Write what this tells you about Problem 9.



two corners at once, yes or no: _____ corner squares: _____ L's: _____

So: _____

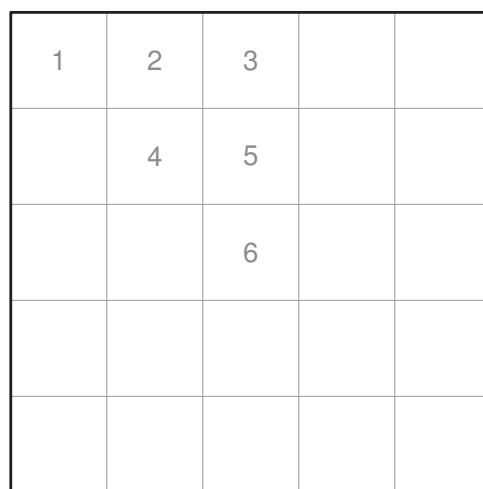
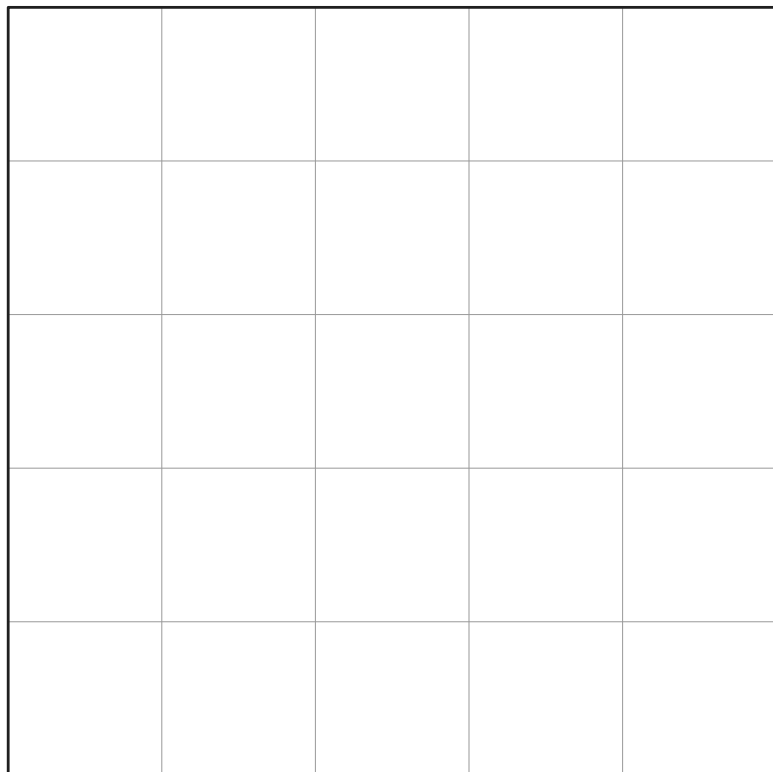
Problem 11: The 8 by 8 board has one black corner. Cover the other 63 squares with 21 L's, using the recipe: the dashed lines cut the board into four 4 by 4 quarters. Put one L in the very middle, covering one square of each quarter that has no black square. Now each quarter is a 4 by 4 board with one square black or covered: do the 4 by 4 recipe inside each quarter. Colour your 21 L's on the board.



Problem 12: A 16 by 16 board with one black square: 255 white squares, 85 L's. Do not build it. Write, in two or three sentences, why the recipe covers it: what the middle L does, and what is left in each quarter. Would the same sentences work for 32 by 32? How many L's would that need?

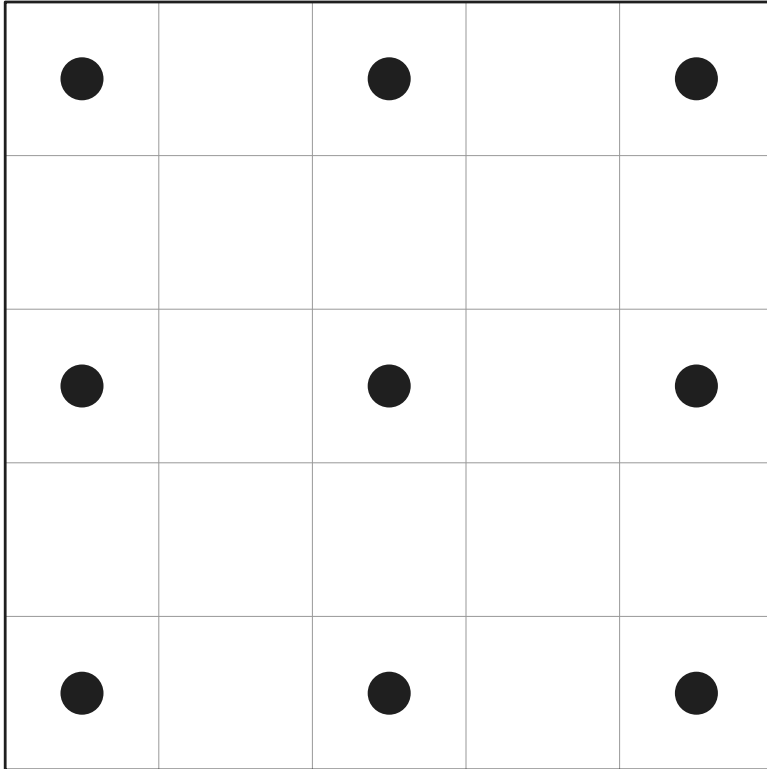
32 by 32: _____ L's: _____

Problem 13: Put the black paper square on the 5 by 5 board at the place marked 1 on the map. Cover the other 24 squares with eight L's. Write Y in square 1 of the map if you could, N if you could not. If you cannot after three different starts, write N. Do the same for places 2 to 6. Then fill in the rest of the map without building: the board looks the same when it is turned or flipped over, so a square that matches a tested square gets the same letter. Describe the pattern of the Y squares in one sentence.



The Y squares are _____

Problem 14: This 5 by 5 board has dots on some squares. Draw a 2 by 2 square anywhere on the board and count the dots inside it; do this in five places. Put an L anywhere on the board: how many dots does it cover? Try ten places and write the most. Eight L's cover at most how many dots? How many dots are on the board? Write, in one sentence, what this says about where the black square must be. Compare with your map from Problem 13.



dots in a 2 by 2 square:

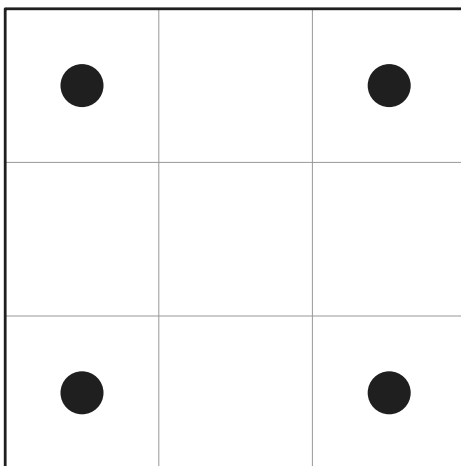
most dots under one L: _____

eight L's cover at most _____ dots

dots on the board: _____

so the black square must be

Problem 15: The 3 by 3 board with dots on its corners. What is the most dots one L can cover? Three L's cover at most how many dots? How many dots are there? Write in one sentence what this says about Problem 9.



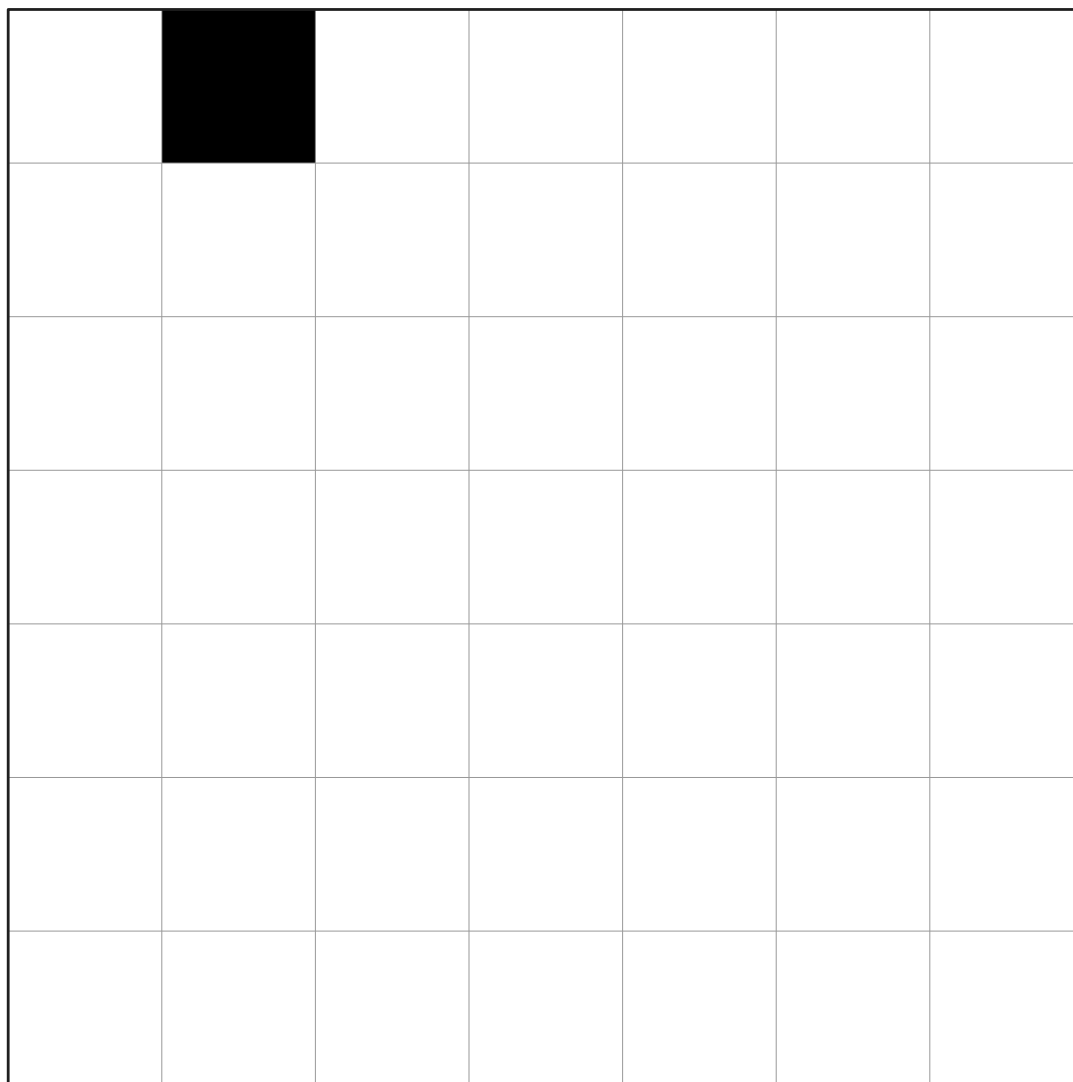
most dots under one L: _____

three L's cover at most _____ dots

dots on the board: _____

so: _____

Problem 16: Before building: mark the dots on this 7 by 7 board yourself, one on every square that is in an odd row and an odd column. Count the dots. Count the L's needed for the 48 white squares. The black square is in row 1, column 2, not on a dot: does the dot argument say N for this board? Then cover the 48 white squares with 16 L's and colour them. Write one sentence about what happened.



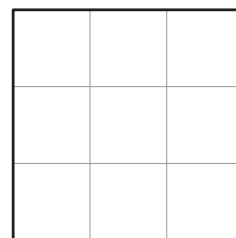
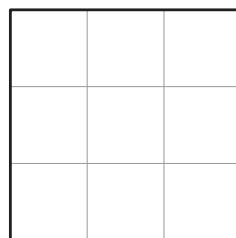
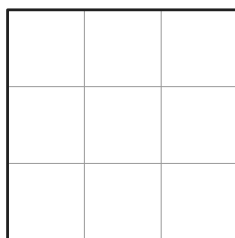
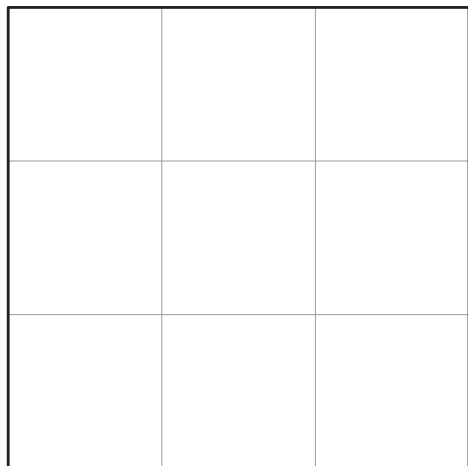
dots: ____

L's: ____

dot argument:

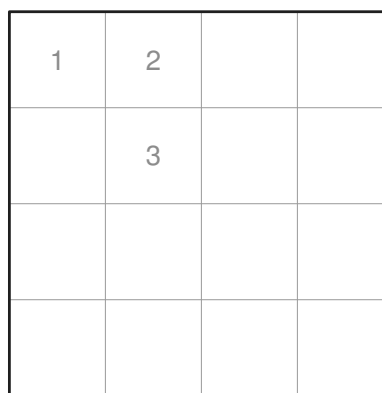
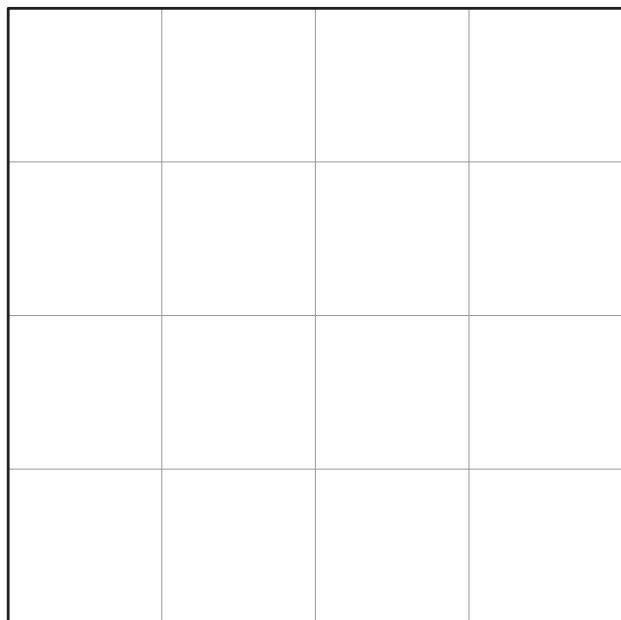
Y or N: _____

Problem 17: A straight tromino covers three squares in a row. Cover the 3 by 3 board with three straight trominoes. Find every way and draw each way on a small board. How many ways?



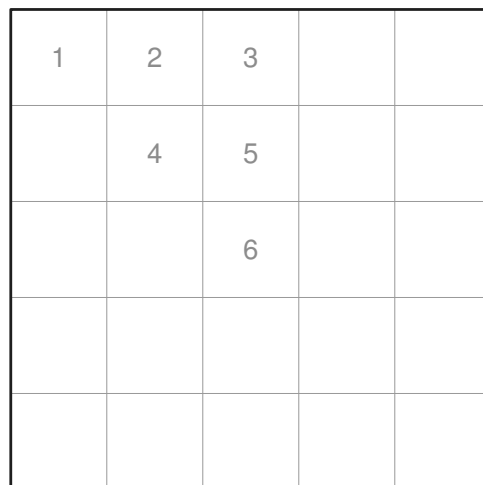
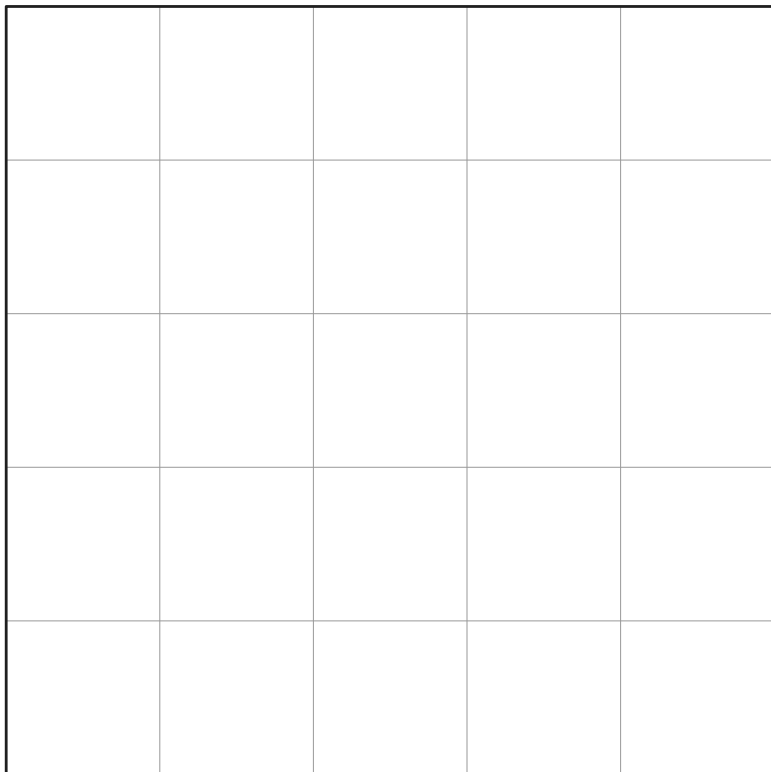
ways: _____

Problem 18: The 4 by 4 board with one black square, to be covered by five straight trominoes. Put the black paper square at place 1 on the map and try to cover the other 15 squares; write Y or N in square 1. If you cannot after three different starts, write N. Do places 2 and 3. Then fill in the rest of the map using the turns and flips of the board. Which squares are Y?



the Y squares: _____

Problem 19: Use the 5 by 5 board below and eight straight trominoes. Put the black square at place 1 on the map, cover the other 24 squares, and write Y or N in square 1. If you cannot after three different starts, write N. Do places 2 to 6, then fill in the rest of the map using the turns and flips of the board. Which squares are Y?



the Y squares: _____

Problem 20: Two ways of numbering the 4 by 4 board with 1, 2, 3 along the diagonals. Put a straight tromino anywhere on the first numbering and read the three numbers under it; try five places, lying (three squares side by side) and standing (three squares one above the other). Count how many 1s, 2s and 3s the first numbering has. How many 1s, how many 2s and how many 3s do five trominoes cover? So which number must the black square carry? Do the same on the second numbering. Which squares carry the right number in both numberings? Compare with your map from Problem 18.

first numbering

1	2	3	1
2	3	1	2
3	1	2	3
1	2	3	1

second numbering

1	3	2	1
2	1	3	2
3	2	1	3
1	3	2	1

under a tromino: _____

first numbering has: 1s ____ 2s ____ 3s ____ five trominoes cover: 1s ____ 2s ____ 3s ____

so the black square must be a ____

second numbering has: 1s ____ 2s ____ 3s ____ five trominoes cover: 1s ____ 2s ____ 3s ____

so the black square must be a ____

right in both numberings: _____

Problem 21: The same two numberings on the 5 by 5 board. Count the 1s, 2s and 3s in each. How many of each do eight trominoes cover? So which number must the black square carry in the first numbering, and in the second? Which squares carry the right number in both? Compare with your map from Problem 19.

first numbering					second numbering				
1	2	3	1	2	2	1	3	2	1
2	3	1	2	3	3	2	1	3	2
3	1	2	3	1	1	3	2	1	3
1	2	3	1	2	2	1	3	2	1
2	3	1	2	3	3	2	1	3	2

first numbering has: 1s ____ 2s ____ 3s ____ eight trominoes cover: 1s ____ 2s ____ 3s ____

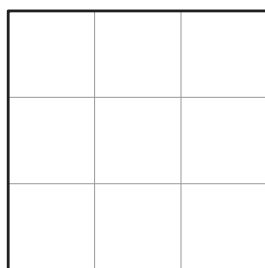
so the black square must be a ____

second numbering has: 1s ____ 2s ____ 3s ____ eight trominoes cover: 1s ____ 2s ____ 3s ____

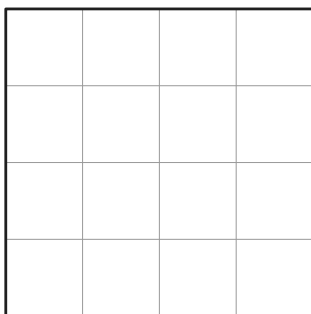
so the black square must be a ____

right in both numberings: _____

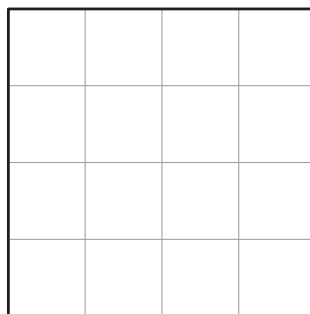
Problem 22: Use the 3 by 3 board from Problem 17 and the 4 by 4 board from Problem 18. Put dominoes on the 3 by 3 board so that no more dominoes can be added anywhere, using as few dominoes as you can. Draw your best answer on the first small board and write the number. Do the same on the 4 by 4 board. Then show that 4 dominoes can never be enough on the 4 by 4: with 4 dominoes, 8 squares stay uncovered and no two of them may be side by side. Where must those 8 squares be? Mark them on the third small board. Then look at the 8 covered squares: can a domino lie on them? Write what you find.



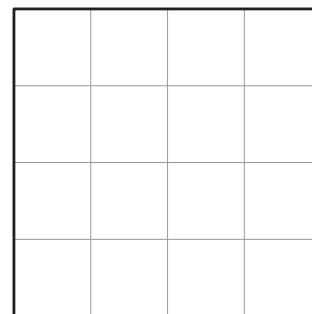
fewest: ____



fewest: ____



with 4 dominoes:



covered squares:

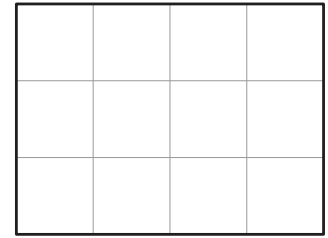
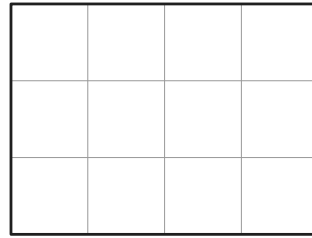
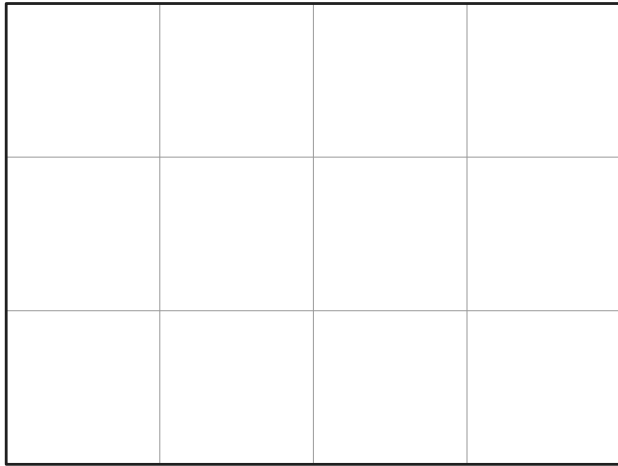
Problem 23: Cover each of the boards 2 by 2, 2 by 3, 2 by 4 and 2 by 5 with dominoes in every possible way, drawing each way on a small board (one thick line through the two squares of each domino; two ways are different if some domino lies in a different place). Write the four counts in the table next to the 2 by 1 count, which is 1, and predict 2 by 6 and 2 by 7. Then look at the left end of each 2 by 5 way: it is either one standing domino (two squares one above the other) or two lying dominoes (two squares side by side). Count each kind, and write what you notice about the two counts.

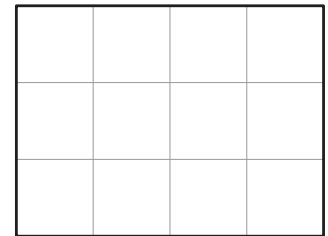
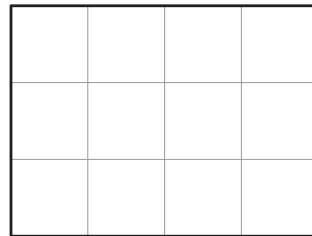
board	2 by 1	2 by 2	2 by 3	2 by 4	2 by 5	2 by 6	2 by 7
ways	1						

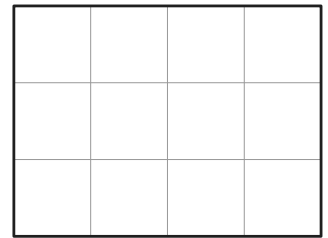
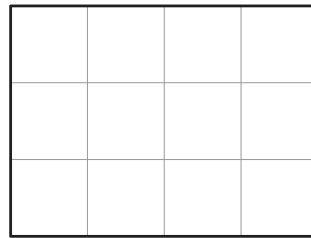
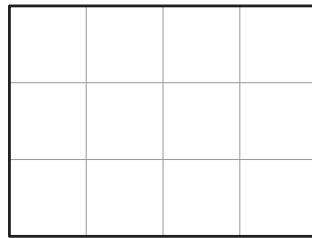
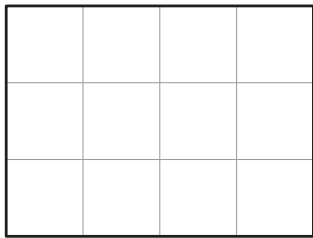
2 by 5, standing at the left end: _____ two lying at the left end: _____

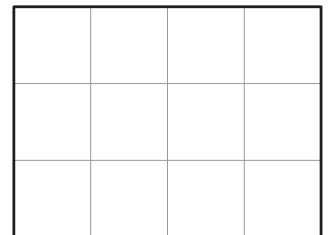
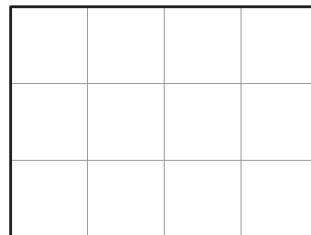
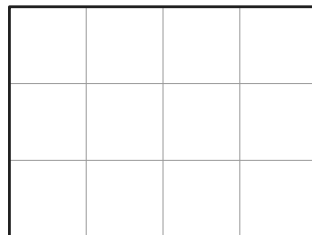
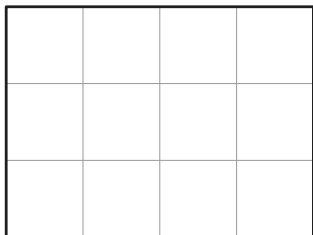
I notice _____

Problem 24: Cover the 3 by 4 board with dominoes in every possible way, drawing each way on a small board. Sort your ways by what covers the top-left square, a lying domino or a standing one, and count each kind. How many ways in all?







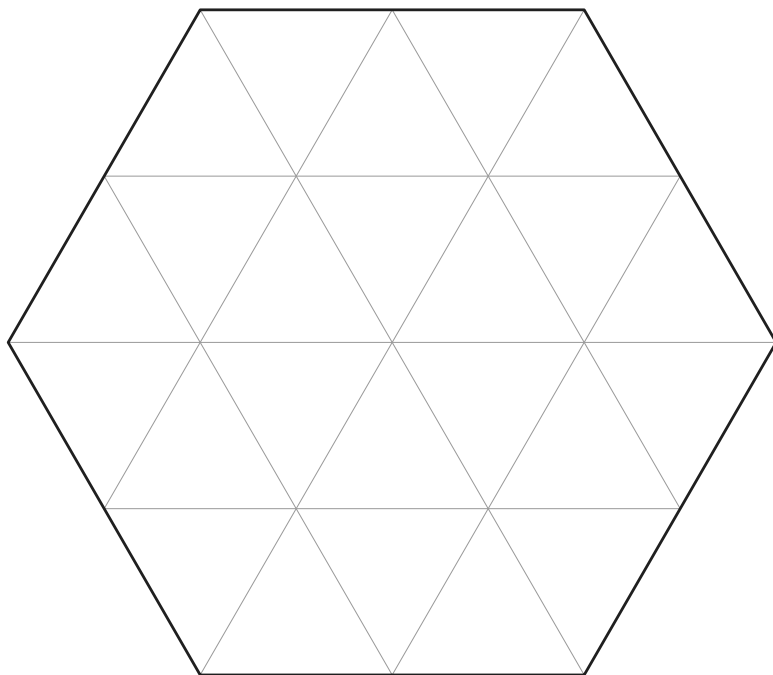


lying at top left: _____ standing at top left: _____ in all: _____

Problem 25: In a 3 by 4 way, find a 2 by 2 block covered by two parallel dominoes. Turning those two dominoes by a quarter turn inside their block is called a flip, and it gives another way. Start from the way with six lying dominoes. On the blank under each of your small boards, write the fewest flips needed to reach that way from the all-lying way. What is the largest number you wrote? Then find two ways that are as far apart as possible: how many flips from one to the other?

largest number of flips from all-lying: _____ farthest two ways: _____ flips apart

Problem 26: Cover the big hexagon with 12 blue rhombi from the pattern blocks. Each rhombus is standing, leaning left or leaning right, as in the three pictures. Count each kind and write the three numbers in the first row of the table. Cover the hexagon in a different way and count again; do four different ways. Draw each way on a small hexagon by drawing the outline of every rhombus.



standing

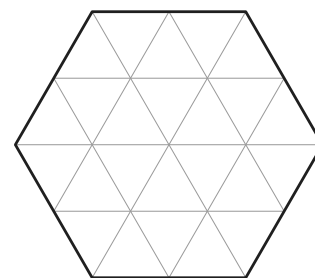
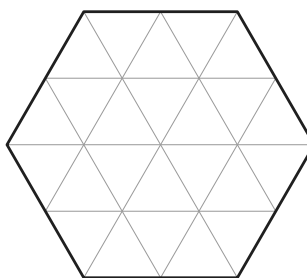
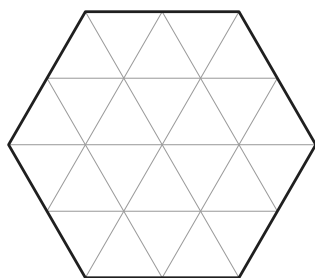
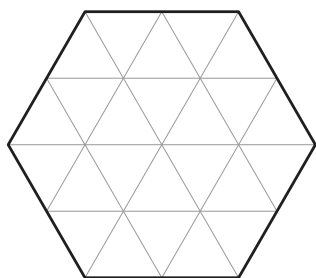


leaning left



leaning right

	standing	left	right
1st way			
2nd way			
3rd way			
4th way			



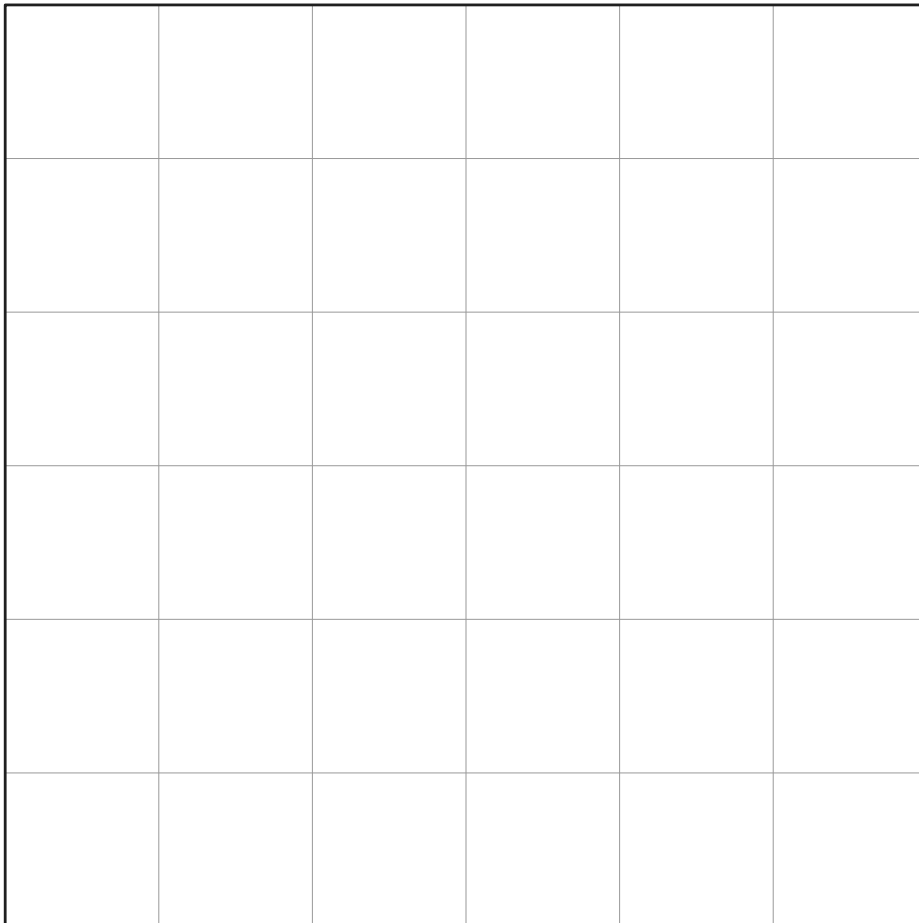
Problem 27: Shade every standing rhombus in one of your drawings from Problem 26. Turn the page a quarter turn to the left: the picture shows a pile of cubes in the corner of a 2 by 2 by 2 box. The way with all shaded rhombi at the bottom is the empty box, 0 cubes; the way with all shaded rhombi at the top is the full box, 8 cubes. How many cubes are in your pile? On the big hexagon, find three rhombi around one point that form a small hexagon and turn the three of them around: that is a spin, and it gives another way. Does a spin add a cube or take one away? Starting from the way with no cubes, what is the fewest spins to reach the way with 8 cubes?

cubes in my drawing: _____

a spin _____

fewest spins from 0 cubes to 8: _____

Problem 28: A straight tetromino covers four squares in a row. Try to cover the 6 by 6 board with nine straight tetrominoes; write Y or N. Then try with nine L-tetrominoes (three squares in a row and one more square beside an end square, making an L; turn and flip it as you like); write Y or N.



straight, Y or N: _____

L-shaped, Y or N: _____

Problem 29: This 6 by 6 board is numbered 1, 2, 3, 4 along the diagonals. Put a straight tetromino anywhere on it, lying (four squares side by side) or standing (four squares one above the other): which numbers are under it? Try five places. Count the 1s, 2s, 3s and 4s on the board. How many of each would nine straight tetrominoes cover? Write what this says about the straight tetrominoes in Problem 28.

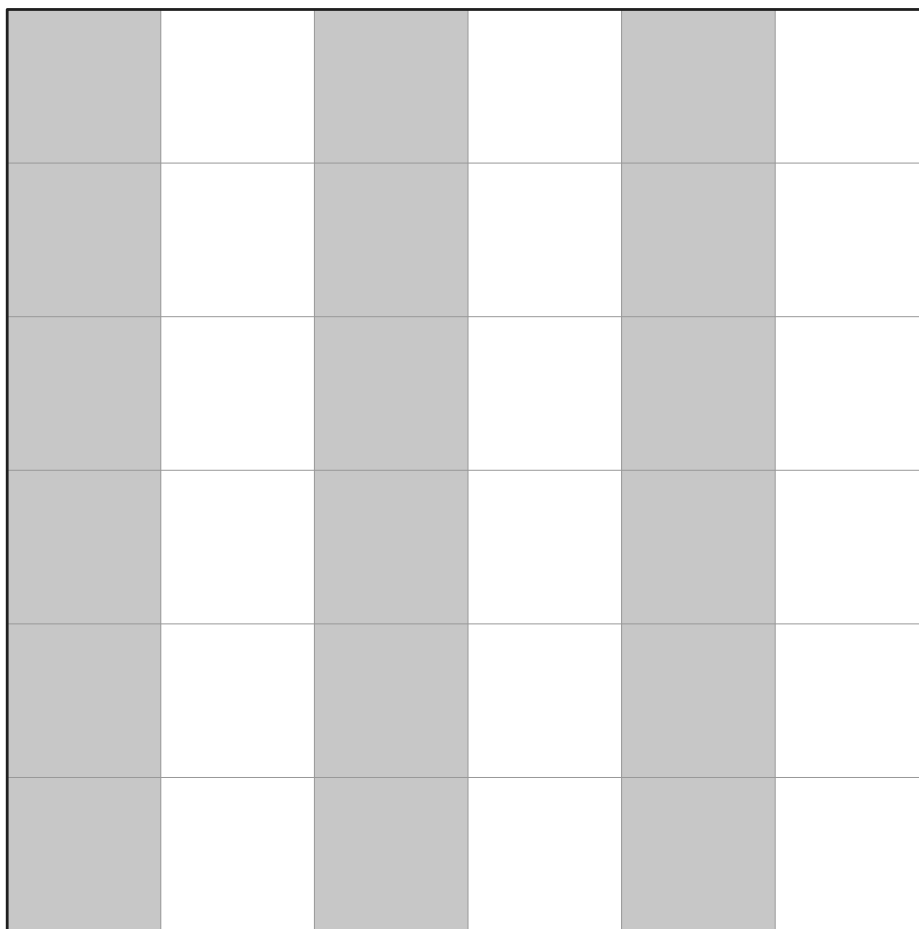
1	2	3	4	1	2
2	3	4	1	2	3
3	4	1	2	3	4
4	1	2	3	4	1
1	2	3	4	1	2
2	3	4	1	2	3

under a straight tetromino: _____

on the board: 1s ____ 2s ____ 3s ____ 4s ____ nine tetrominoes cover: 1s ____ 2s ____ 3s ____ 4s ____

so: _____

Problem 30: The same 6 by 6 board with every other column shaded. Put an L-tetromino anywhere on it: how many shaded squares does it cover? Try eight places, turned and flipped. Nine L-tetrominoes together cover an odd or an even number of shaded squares? How many shaded squares are on the board? Write what this says about the L-tetrominoes in Problem 28.



shaded squares under an L-tetromino: _____

nine of them cover an _____ number of shaded squares shaded squares on the board: _____

so: _____